HW5

6.1

> Construct an E-R diagram for a car insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents. Each insurance policy covers one or more cars and has one or more premium payments associated with it. Each payment is for a particular period of time, and has an associated due date, and the date when the payment was received.

One possible ER diagram is shown in Figure 6.101. Payments are modeled as weak entities since they are related to a specific policy. Note that the participation of accident in the relationship *participated* is not total, since it is possible that there is an accident report where the participating car is unknown.

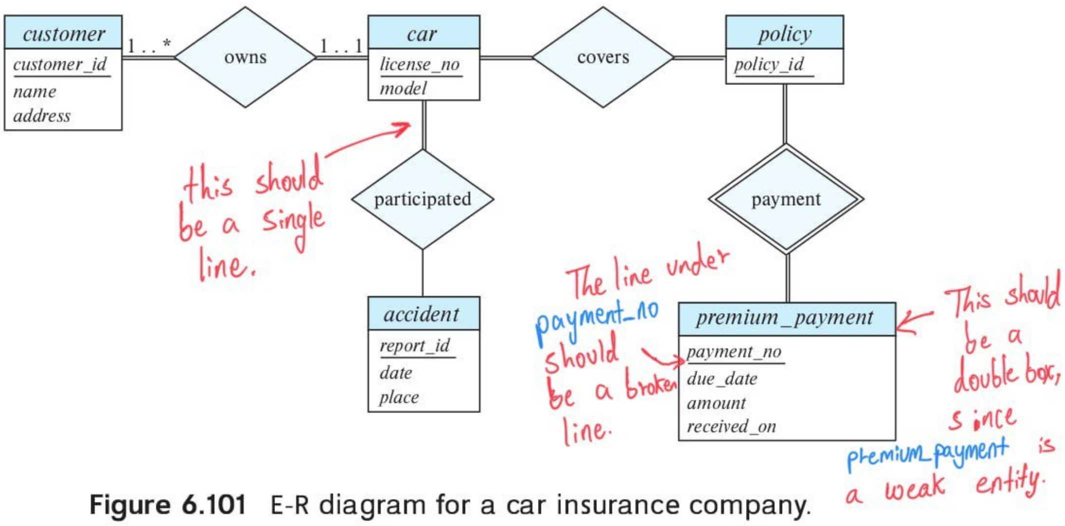


Figure 6.101 E-R diagram for a car insurance company.

Note that the double-line connecting the entity set *car* to the relationship *participated* should be a single-line, since we could have cars that have not participated in any accident until now. That is, the participation of the entity *car* in the relationship *participated* is **not total**, thus it is **partial**.

Also note that, given a *premium_payment* we can not identify for which *car* that *premium_payment* was made for. So there is some bug that needs to get fixed in the E-R diagram if we want to determine the *car* given a *premium_payment* info.

6.2

> Consider a database that includes the entity sets *student*, *course*, and *section* from the university schema and that additionally records the marks that students receive in different exams of different sections.
>
> a. Construct an E-R diagram that models exams as entities and uses a ternary relationship as part of the design.
>
> b. Construct an alternative E-R diagram that uses only a binary relationship between *student* and *section*. Make sure that only one relationship exists between a particular *student* and *section* pair, yet you can represent the marks that a student gets in different exams.

a. The E-R diagram is shown in Figure 6.102. Note that an alternative is to model examinations as weak entities related to a section, rather than as strong entities. The *exam_marks* relationship would then be a binary relationship between *student* and *exam*, without directly involving *section*.

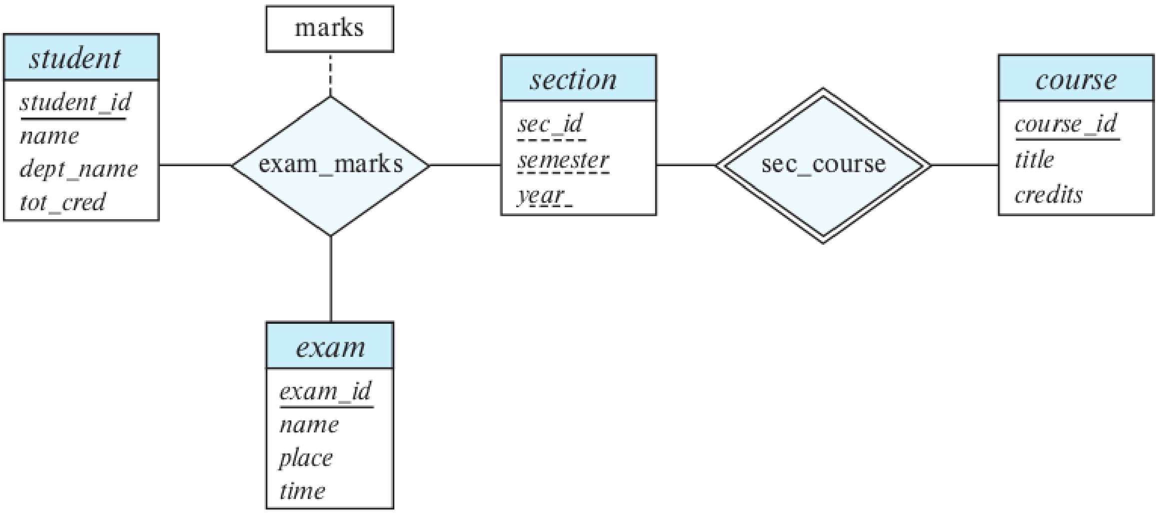


Figure 6.102 E-R diagram for marks database.

b. The E-R diagram is shown in Figure 6.103. Note that here we have not modeled the name, place, and time of the exam as part of the relationship attributes. Doing so would result in duplication of the information, once per student, and we would not be able to record this information without an associated student. If we wish to represent this information, we need to retain a separate entity corresponding to each exam.

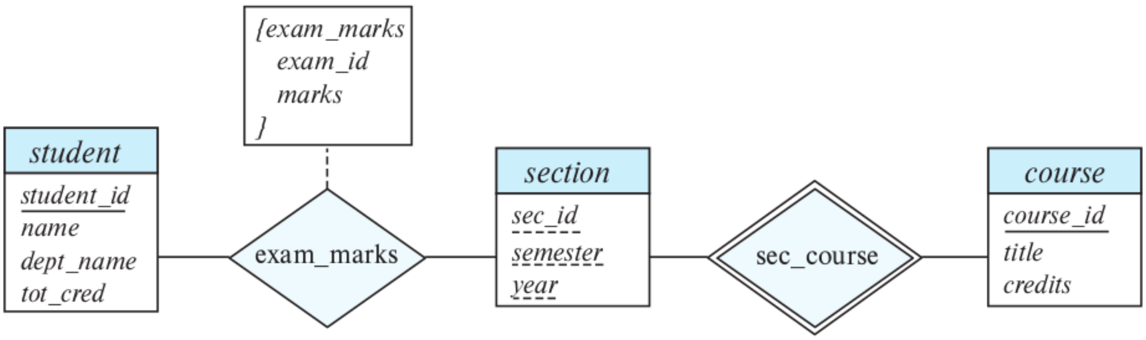


Figure 6.103 Another E-R diagram for marks database.

Note that: *exam_marks*, the *descriptive attribute* of the relationship *exam_marks*, is a composite multivalued attribute.

6.21

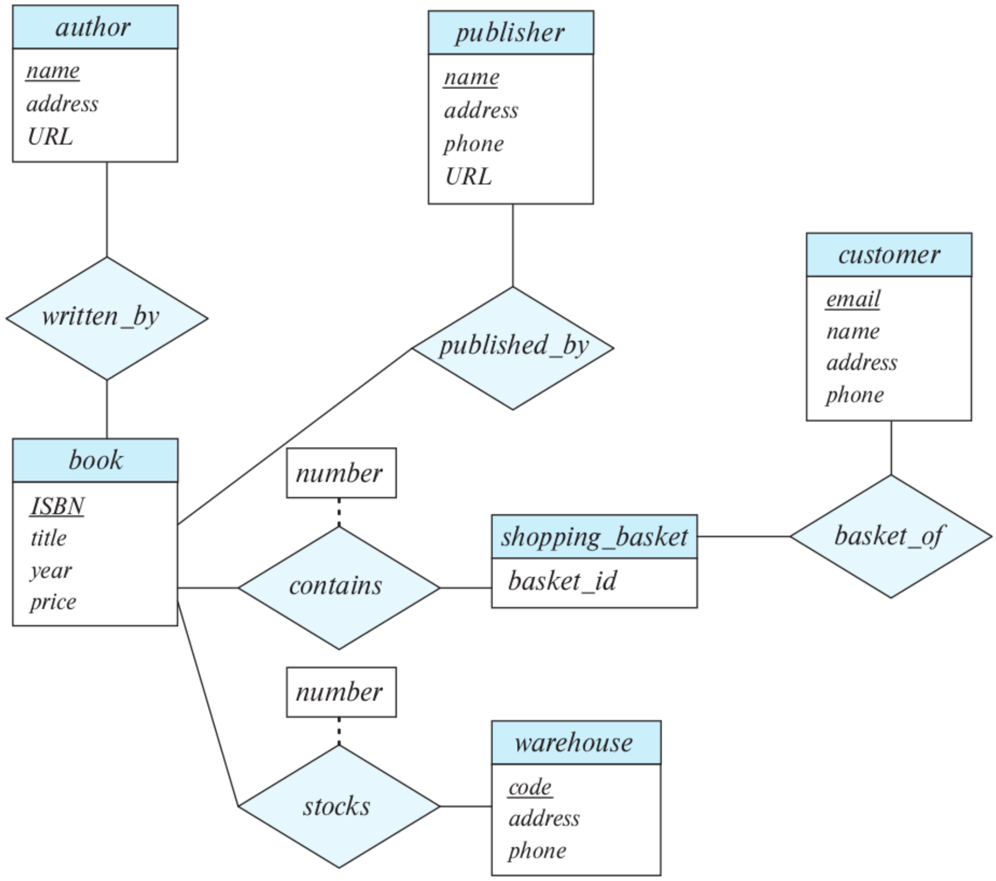


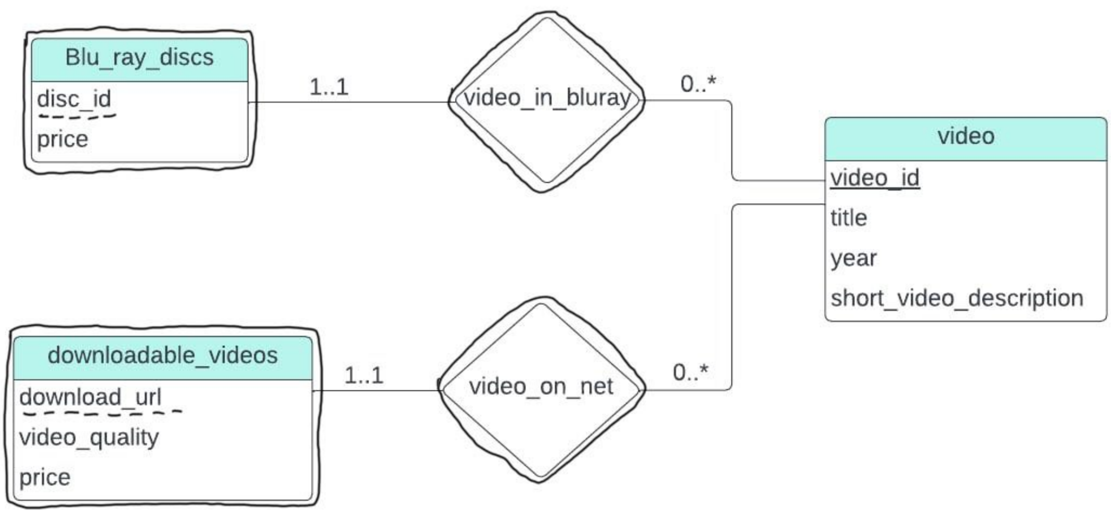
Figure 6.30 E-R diagram for modeling an online bookstore.

Consider the E-R diagram in Figure 6.30, which models an online bookstore.

a. Suppose the bookstore adds Blu-ray discs and downloadable video to its collection. The same item may be present in one or both formats, with differing prices. Draw the part of the E-R diagram that models this addition, show just the parts related to video.

b. Now extend the full E-R diagram to model the case where a shopping basket may contain any combination of books, Blu-ray discs, or downloadable video.

a.



Note that `Blu_ray_discs` and `downloadable_videos` are weak entities while `video_in_bluray` and `video_on_net` are the identifying relationships sets. `video` is the identifying entity set and owns both of the weak entities.

b.

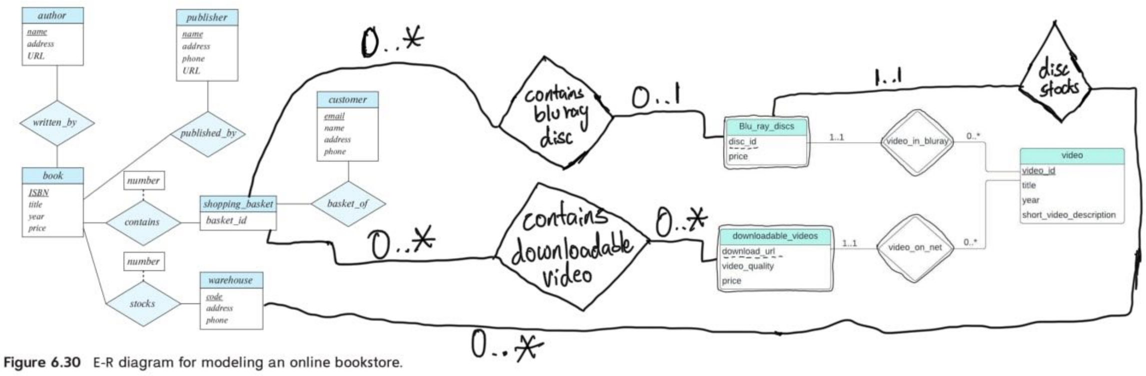
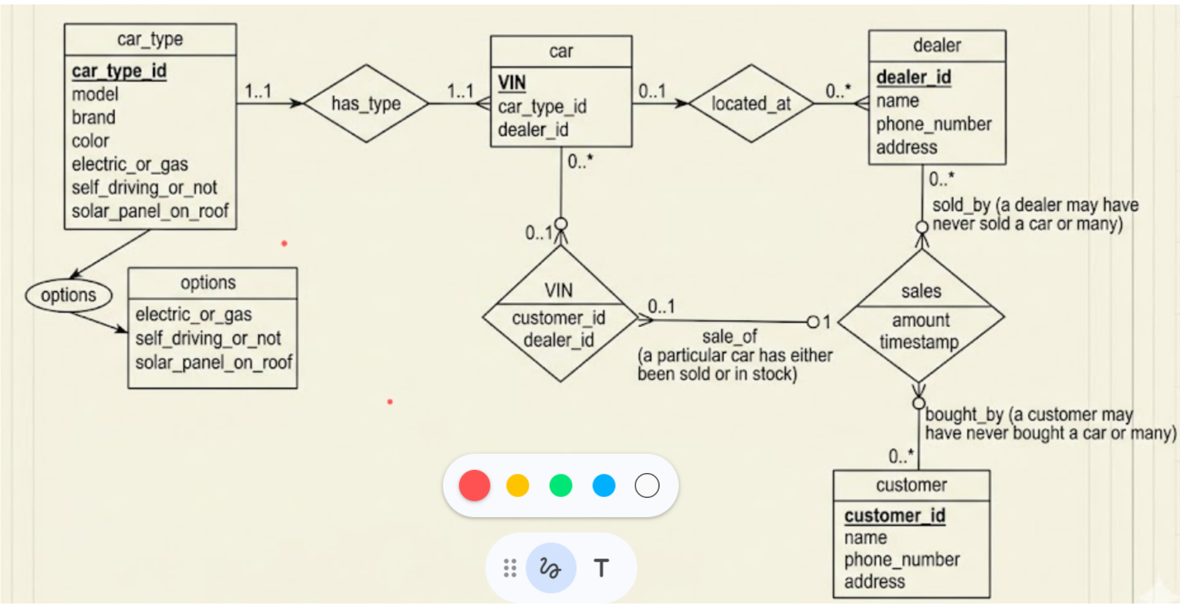


Figure 6.30 E-R diagram for modeling an online bookstore.

6.22

> Design a database for an automobile company to provide to its dealers to assist them in maintaining customer records and dealer inventory and to assist sales staff in ordering cars.
>
> Each vehicle is identified by a vehicle identification number (VIN). Each individual vehicle is a particular model of a particular brand offered by the company (e.g., the XF is a model of the car brand Jaguar of Tata Motors). Each model can be offered with a variety of options, but an individual car may have only some (or none) of the available options. The database needs to store information about models, brands, and options, as well as information about individual dealers, customers, and cars.
>
> Your design should include an E-R diagram, a set of relational schemas, and a list of constraints, including primary-key and foreign-key constraints.

The following figure displays the E-R diagram of the database for the automobile company. The attribute `options` of the entity set `car_type` is a composite attribute. The ternary relationship set `sales` represents a single trasaction or a sale of a car. A `dealer` may have never sold a car or have sold numerous cars. A `customer` may have never bought a car or have bought numerous cars. But a particular car has either been sold or in stock. This constraints are represented as mapping cardinalities in the diagram.



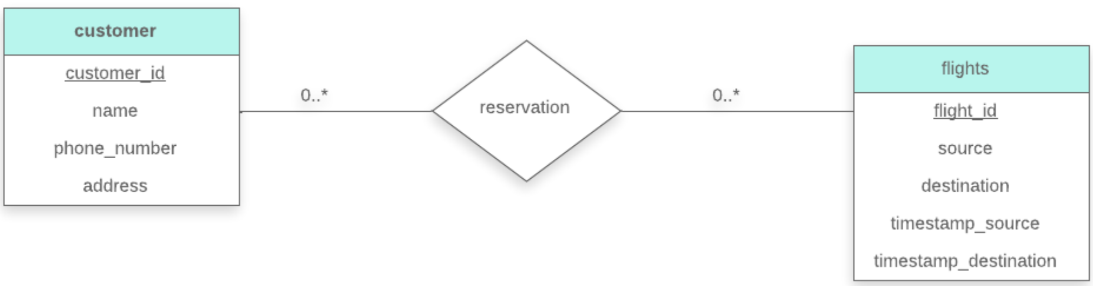
When we change the diagram to a relational schema we get the following:

```
car_type(car_type_id, model, brand, color, electric_or_gas,
self_driving_or_not,solar_panel_on_roof)
car(VIN, car_type_id, dealer_id)
customer(customer_id, name, phone_number, address)
dealer(dealer_id, name, phone_number, address)
sale(VIN, customer_id, dealer_id, amount, timestamp)
```

The attributes `car_type_id` and `dealer_id` in the relation `car` are foreign-keys referencing `car_type` and `dealer` relations respectively. The `sale` relation has `VIN`, `customer_id` and `dealer_id` as foreign-keys referencing the relations `car_type`, `customer` and `dealer` respectively.

6.24

> Design a database for an airline. The database must keep track of customers and their reservations, flights and their status, seat assignments on individual flights, and the schedule and routing of future flights.
>
> Your design should include an E-R diagram, a set of relational schemas, and a list of constraints, including primary-key and foreign-key cosntraints.



Relation schemas:

```
customer(customer_id, name, phone_number, address)
flights(flight_id,src,dest,timestamp_src,timestamp_dest)
reservation(customer_id,flight_id)
```